SQL Saturday #176

Pordenone, Italy

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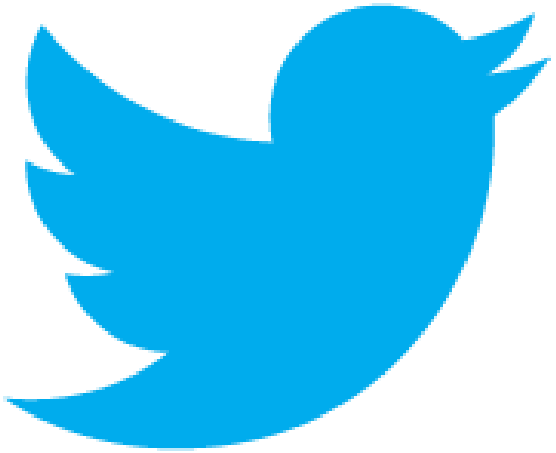

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Speaker

▶ Gianluca Hotz

- Consultant and mentor at SolidQ
  - Administration, capacity planning, data modeling, development, optimization
  - >15 years on SQL Server (from 4.21 in 1996)
- Interests
  - Relational model, DBMS architectures, high availability and disaster recovery
- Microsoft MVP SQL Server since 1998
- Founder & vice-president UGISS
  - User Group Italiano SQL Server
- Mail: ghotz@ugiss.org - ghotz@solidq.com

AlwaysOn

▶ Introduction

- New name for HA and DR technologies
  - Failover Cluster Instance (FCI)
  - Availability Groups (AG)
- Previously named HADRON
  - Keep in mind when searching articles
  - Still present in some DMVs (_hadr_)
- For the purpose of this presentation
  - High Availability (local HA): in a data center
  - Disaster Recovery: across data centers

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▶ Brief review

- Protects an entire SQL Server instance
- Requires Shared Storage
- Geographic DR with SQL Server 2008
  - Proprietary solutions for Stretch V-LAN
  - Proprietary solutions for storage replication

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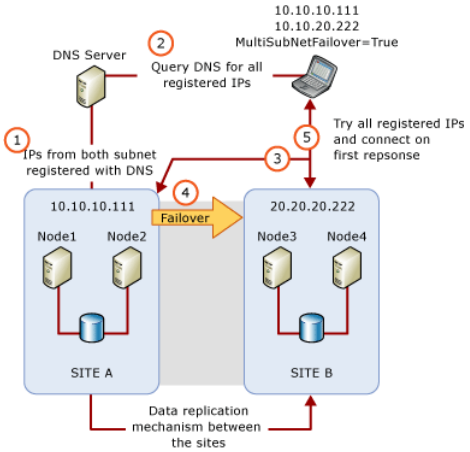
▶ Multi-Subnet Support

- Nodes on different subnets
  - IP Address resources in OR
  - Automatic DNS updates
- How do clients reconnect?
  - New connection string parameter
    - MultiSubnetFailover=True
  - For old clients
    - Reduce timeout value
    - Reduce DNS updates latency (HostRecordTTL)
    - Reduce DNS cache expiration on the client
- Still need storage replication technology



http://msdn.microsoft.com/en-us/library/ff878716.aspx

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▶ Failover policies in SQL Server 2008

- Looks Alive
  - Checks service state (Service Control Mgr.)
  - Every 5 seconds
- IsAlive
  - Checks SELECT @@SERVERNAME
  - Every 60 seconds
  - Can pass check even when not responding to other incoming requests
    - e.g. stuck schedulers

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▶ Flexible failover policies in SQL Server 2012

- Resource DLL calls system procedure
  - **sp_server_diagnostic**
- Looks Alive, IsAlive based on
  - Diagnostic information from procedure
    - New instance parameter **FailureConditionLevel**
    - Allows selecting conditions for failover to occur
  - Timeouts while calling diagnostic procedure
    - New Cluster parameter **HealthCheckTimeout**

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▶ Failure condition levels

Level	Condition	Description
0	No automatic failover or restart	Never Failover (system maintenance only)
1	Failover or restart on SQL Server down	Service down
2	Failover or restart on SQL Server unresponsive	Server hanged (sp not responding within HealthCheckTimeout)
3	Failover or restart on critical SQL Server errors	System unhealthy (diagnostics sp returns “system error”)
4	Failover or restart on moderate SQL Server errors	Resource unhealthy (sp returns “resource error”)
5	Failover or restart on any qualified failure	QP Unhealthy (sp returns “query_processing error”)

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▶ Flexible failover policies configuration

- Failover Cluster Manager
- PowerShell
- T-SQL!
  - ALTER SERVER CONFIGURATION FAILOVER CLUSTER PROPERTY HEALTHCHECKTIMEOUT = 60000;
  - ALTER SERVER CONFIGURATION SET FAILOVER CLUSTER PROPERTY FAILURECONDITIONLEVEL = 3;

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▶ Reducing planned downtime with SQL Server 2012

- Windows Core Support
  - Less patch to be installed (circa 50-60%)
- Support for rolling upgrade and patching
- Better control on failover time
  - Indirect Checkpoints
  - ALTER DATABASE …
    SET TARGET_RECOVERY_TIME = …

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▶ New Storage options with SQL Server 2012

- Tempdb on local storage
  - Can leverage fast SSD cards
- SMB shares supported
  - Best on
    - Windows Server 2008 R2 (SMB 2.1 + Hotfix)
    - Windows Server 2012
- Asymmetric Disks

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▶ Data replicas review

- Log Shipping
- Database Mirroring
- Replication
- SQL Server 2012: Availability Groups

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▶ Common Problems

- Manual Failover*
- No coordination to failover multiple DBs
- No backups on replicas*
- Specific limitations
  - E.g. FILESTREAM, cross-database consistency of transactions

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▶ AG as evolution of Database Mirroring

- Coordinated failover of multiple DBs
- Multiple replicas (up to 4)
- Synchronous/asynchronous replicas
- Up-to-date readable replicas
- Ability to off-load maintenance on replicas e.g. backups

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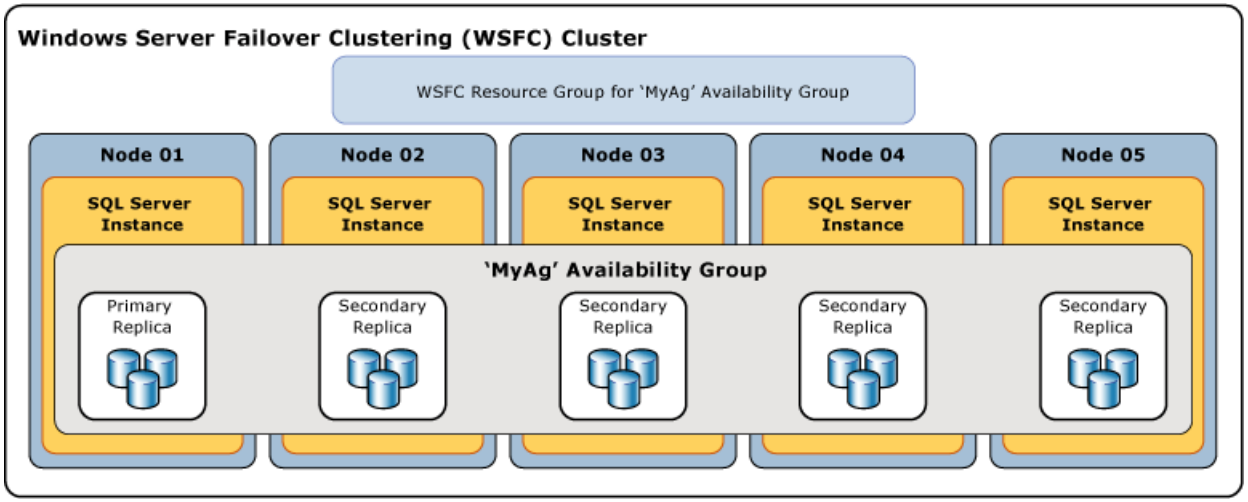
▶ Overview



http://msdn.microsoft.com/en-us/library/ff877884.aspx

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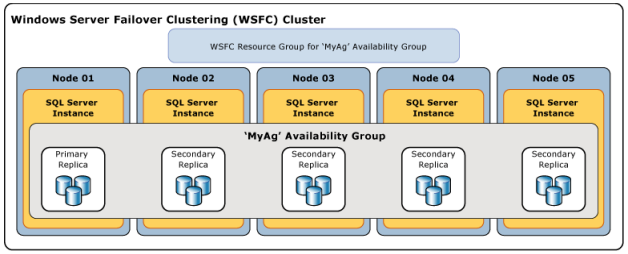
▶ Windows Server Failover Clustering (WSFC)



- Used by Availability Groups for
  - Primary health detection
  - Inter-node health detection
  - Failover coordination
  - Distributed data store for settings and state
  - Distributed change notifications between nodes

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▶ Synchronization and Failover

- Availability Mode
  - Synchronous
  - Asynchronous
- Failover mode
  - Automatic
  - Manual
  - Forced

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▶ Conditions for Automatic Failover

- Primary and Secondary need
  - Synchronous Availability Mode configuration
  - Automatic Failover configuration
  - To be synchronized
- WFSC needs to reach a quorum
- Failover policies met

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▶ Flexible failover policies for Availability Groups

- Leverages FCI mechanism
  - **sp_server_diagnostic**
- Configuration in T-SQL
  - **ALTER AVAILABILITY GROUP …**
    SET (HEALTH_CHECK_TIMEOUT = 30000);
  - **ALTER AVAILABILITY GROUP …**
    SET (FAILURE_CONDITION_LEVEL = 3);

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▶ Replicas and backups

- Backup support on secondary replicas
  - Full backups with **COPY_ONLY**
  - Differential not supported!
  - Log backups
    - Form a single log-chain across all replicas
- Automated Backup Preference & Priority
  - No immediate effect
  - Backup scripts need to query UDF
    - **sys.fn_hadr_backup_is_preferred_replica('dbname')**

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▶ Read-Only Access

- Replicas roles
  - Primary: **READ_WRITE | ALL**
  - Secondary: **NO | READ_ONLY | ALL**
- New connection property
  - **ApplicationIntent**
  - Name slightly different across client APIs
- Read-Only Routing

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▶ Topology example 1

Direct attached storage local, regional and geo secondaries

Synchronous data movement

Asynchronous data movement

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▶ Demo

- Primary Datacenter
  - Primary replica
  - Secondary replica for HA and reporting
- Secondary Datacenter
  - Secondary replica for DR

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▶ Read-Only Access Capacity Planning

- Synchronization latency of replica can raise because of the read I/O activity
- Impact on tempdb
  - Temporary statistics
    - Automatically created
    - Gone after failover
  - Read Committed Snapshot Isolation
    - 14 bytes added to rows at primary (on updates)

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▶ Problem with replicas at database level

- Need Manual synchronization for objects outside database scope
  - New logins/users
  - Jobs
- Solution for logins/users
  - Contained databases
  - Problem with **ApplicationIntent=ReadOnly**

Advanced AlwaysOn Scenarios

▶ WSFC: Quorum, votes and DR site

- Quorum
  - Majority Node, Majority Node and Fileshare
  - Can’t use Quorum Disk without shared storage
- Windows 2008 e 2008 R2 + QFE
  - Allow to assign voting rights to nodes
  - Avoid scenarios where DR site takes over
  - Common scenario is to always failover manually to DR site i.e. DR site is not considered for HA

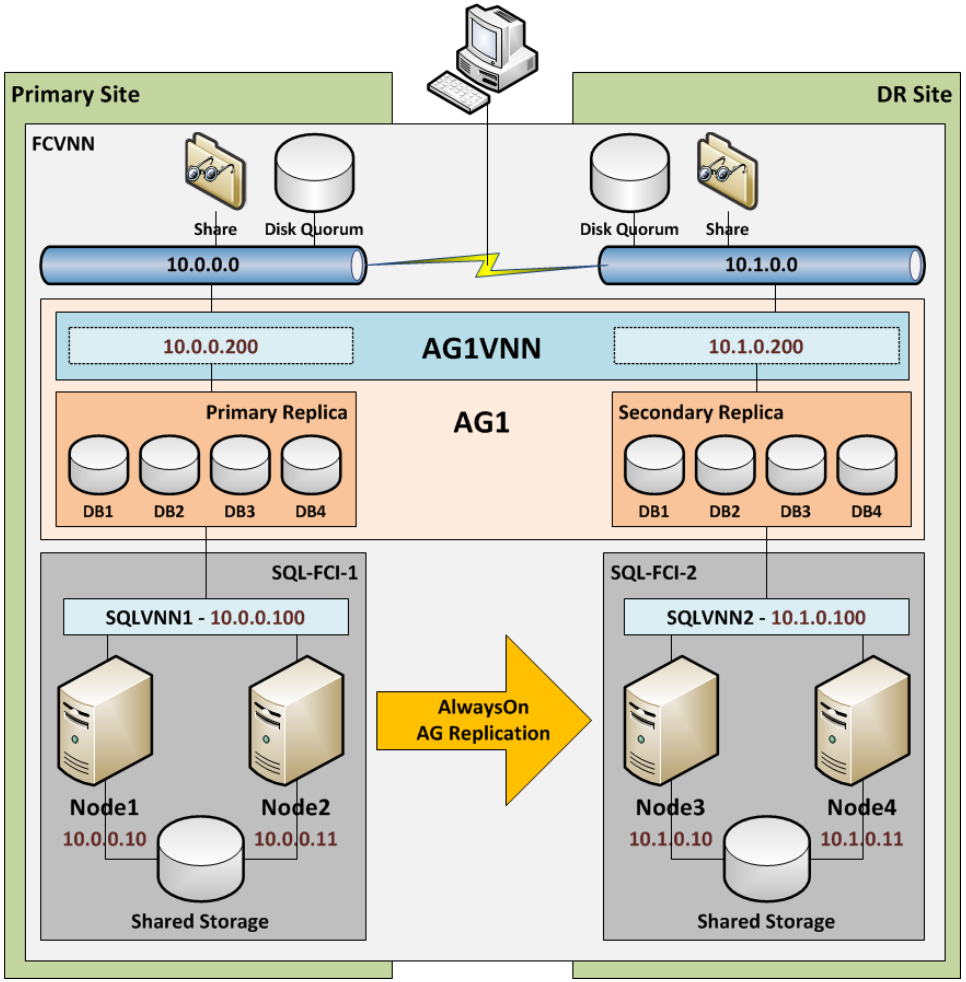
Advanced AlwaysOn Scenarios

▶ WSFC: Asymmetric Storage

- Windows 2008 + QFE or 2008 R2 + SP1
- Disks shared only among some nodes
  - E.g. one SAN on primary site and one SAN on DR site, volumes shared only among local nodes while the cluster stretches across sites
- Allows new Quorum models
  - Asymmetric Disk Majority
  - Node Majority + Asymmetric Disk Majority

Advanced AlwaysOn Scenarios

- Multi-site FCI+AG scenario



Advanced AlwaysOn Scenarios

▶ Topology Example 2

Shared storage, regional and geo secondaries

Asynchronous data movement

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HA/DR Editions and Support

▶ Minimum Support for non Enterprise editions

- Server Core
- Log Shipping (except on Express)
- Database Mirroring
  - Deprecated!
  - Synchronous only for BI/Standard
  - Witness only for Web/Express
- Failover Clustering Instance
  - 2 nodes for BI/Standard
- Purely passive nodes licensing as in 2008

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SQL Server 2012 Service Pack 1

▶ AlwaysOn related fixes

- Fixes here
  - http://support.microsoft.com/kb/2674319
- CU3 and CU4 fixes not included
  - http://support.microsoft.com/kb/2723749
  - http://support.microsoft.com/kb/2758687
- Problems with Slipstream ISO images
  - http://support.microsoft.com/kb/2783963

SQL Server 2012 Service Pack 1

▶ AlwaysOn new features

- Cross-Cluster Migration of AlwaysOn Availability Groups for OS Upgrade
  - E.g. upgrading to Windows 2012
  - Books Online
    - http://msdn.microsoft.com/en-us/library/cc645581.aspx
  - Whitepaper
    - http://go.microsoft.com/fwlink/?LinkId=265471
    - Still missing as of today!

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▶ AG Main Resources

- Books Online
- SQL AlwaysOn Blog
  - http://blogs.msdn.com/b/sqlalwayson
- CSS SQL Server Engineers
  - http://blogs.msdn.com/b/psssql/archive/tags/alwayson
- SQL CAT
  - http://sqlcat.com

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▶ AG Whitepapers series: AlwaysOn HA/DR architectures and design patterns

- **Multisite Failover Cluster Instance**
  - http://sqlcat.com/sqlcat/b/whitepapers/archive/2011/12/22/sql-server-2012-alwayson_3a00_-multisite-failover-cluster-instance.aspx
- **SQL Server 2012 AlwaysOn High Availability and Disaster Recovery Design Patterns**
  - http://sqlcat.com/sqlcat/b/msdnmirror/archive/2011/12/22/sql-server-2012-alwayson-high-availability-and-disaster-recovery-design-patterns.aspx
- **Building a High Availability and Disaster Recovery Solution by Using AlwaysOn Availability Groups**
  - http://sqlcat.com/sqlcat/b/whitepapers/archive/2012/06/22/alwayson-architecture-guide-building-a-high-availability-and-disaster-recovery-solution-by-using-alwayson-availability-groups.aspx
- **Building a High Availability and Disaster Recovery Solution by Using Failover Cluster Instances and Availability Groups**
  - http://sqlcat.com/sqlcat/b/whitepapers/archive/2012/06/22/alwayson-architecture-guide-building-a-high-availability-and-disaster-recovery-solution-by-using-failover-cluster-instances-and-availability-groups.aspx

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▶ AG Resources: articles

- **Dashboard**
  - http://www.sqlskills.com/blogs/joe/post/Answering-Questions-with-the-AlwaysOn-Dashboard.aspx
- **Health Model**
  - http://blogs.msdn.com/b/sqlalwayson/archive/2012/02/09/overview-of-the-alwayson-manageability-health-model.aspx
  - http://blogs.msdn.com/b/sqlalwayson/archive/2012/02/13/extending-the-alwayson-health-model.aspx
- **Client Reconnect**
  - http://www.mattmasson.com/index.php/2012/03/alwayson-multi-subnet-failover-and-ssis
- **Failing Over Stack Overflow and Stack Exchange for Sandy**
  - http://blog.serverfault.com/2012/10/30/failing-over-stack-overflow-and-stack-exchange-for-sandy
- **Rick's SQL Server Blog**
  - http://dangerousdba.blogspot.com

Questions & Answers

GRAZIE!

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